

- If the sets $A = \{-2, 0, 2\}$ and $B = \{0\}$, then ()
 A. $A \in B$ B. $B \in A$ C. $A \subseteq B$ D. $B \subseteq A$
- The solution set of the inequality $x^2 - 3x + 2 < 0$ is ()
 A. $(-\infty, 1)$ B. $(2, +\infty)$ C. $(1, 2)$ D. $(-\infty, 1) \cup (2, +\infty)$
- The domain of the function $f(x) = \sqrt{x-1} + \frac{1}{x-2}$ is ()
 A. $(1, +\infty)$ B. $[1, +\infty)$
 C. $[1, 2)$ D. $[1, 2) \cup (2, +\infty)$
- In the arithmetic sequence $\{a_n\}$, where $a_1 = 1$ and the common difference is $d = 2$, the value of a_8 is ()
 A. 13 B. 14 C. 15 D. 16
- In the Cartesian coordinate system, the quadrant in which the point $P(x^2 + 2, -3)$ lies is ()
 A. First quadrant B. Second quadrant
 C. Third quadrant D. Fourth quadrant
- Given $\sin\left(\alpha + \frac{\pi}{6}\right) = \frac{4}{5}$, then $\cos\left(\frac{\pi}{3} - \alpha\right) =$ ()
 A. $-\frac{4}{5}$ B. $-\frac{3}{5}$ C. $\frac{3}{5}$ D. $\frac{4}{5}$
- Which of the following functions is odd? ()
 A. $y = x^3 + x$ B. $y = x^3 + x^2$ C. $y = \frac{1}{x-1}$ D. $y = x^2 + 6$
- The inverse function of $y = \log_3 x$ is ()
 A. $y = 3^x$ B. $y = x^3$ C. $y = \left(\frac{1}{3}\right)^x$ D. $y = \log_{\frac{1}{3}} x$
- Among the following functions, the one that is monotonically increasing over its domain is ()
 A. $y = \left(\frac{1}{2}\right)^x$ B. $y = \frac{1}{x}$ C. $y = \tan x$ D. $y = \log_{\sqrt{2}} x$
- In the Cartesian coordinate system, point M lies in the fourth quadrant. Its distances from the x -axis and y -axis are 6 and 4, respectively. The coordinates of point M are ()

- A. $(4, -6)$ B. $(-4, -6)$ C. $(6, -4)$ D. $(-6, -4)$
11. The solution to the fractional inequality $\frac{x+2}{x-1} \geq 2$ is ()
- A. $1 \leq x \leq 4$ B. $1 < x < 4$ C. $1 < x \leq 4$ D. $1 \leq x < 4$
12. In the arithmetic sequence $\{a_n\}$, $a_3 + a_7 = 10$, $a_5 =$ ()
- A. 5 B. 6 C. 7 D. 8
13. The slope of the line $y = x - 3$ is ()
- A. 30° B. 45°
C. 135° D. 150°
14. The following inequality holds true ()
- A. $\left(\frac{3}{4}\right)^{\frac{2}{3}} < \left(\frac{3}{4}\right)^{\frac{3}{4}}$ B. $\log_2 5 < \log_4 12$ C. $\log_7 3 > \frac{\sqrt{5}}{5}$ D. $(\sqrt{2})^{3.9} > 3.9$
15. Which of the following pairs of values is correctly ordered? ()
- A. $3^{0.8} < 3^{0.7}$ B. $0.75^{-0.1} < 0.75^{0.1}$
C. $\log_2 3.4 < \log_2 8.5$ D. $\log_{0.3} 1.8 < \log_{0.3} 2.7$
16. In the Cartesian coordinate system, point $P(-1, 3)$ lies on the terminal side of angle α . Then, $\tan 2\alpha =$ ()
- A. $-\frac{4}{3}$ B. $\frac{3}{4}$ C. $-\frac{3}{5}$ D. $\frac{3}{5}$
17. For an arithmetic sequence $\{a_n\}$ satisfying $a_1 = 1, a_3 = 3$, then $a_{2024} =$ ()
- A. 2024 B. 2025 C. 2026 D. 2027
18. If circle C is tangent to the axis x and its center coordinates are $(1, 2)$, then the equation of circle C is ()
- A. $x^2 + y^2 - 2x - 4y + 1 = 0$ B. $x^2 + y^2 - 2x - 4y - 1 = 0$
C. $x^2 + y^2 - 2x - 4y - 3 = 0$ D. $x^2 + y^2 - 2x - 4y + 3 = 0$
19. Given the equation of circle M as $(x+1)^2 + (y-2)^2 = 4$, the coordinates of the center M are ()
- A. $(-1, 2)$ B. $(1, 2)$ C. $(1, -2)$ D. $(-1, -2)$

20. The equation of the line passing through point $(1,1)$ with normal vector $(2,1)$ is ()
- A. $2x - y + 1 = 0$ B. $2x + y - 3 = 0$ C. $x - 2y + 1 = 0$ D. $x + 2y - 3 = 0$
21. The equation of the line passing through the point $(3,1)$ with slope $\frac{1}{2}$ is ()
- A. $x - 2y - 1 = 0$ B. $x + 2y - 5 = 0$
C. $2x - y - 5 = 0$ D. $2x + y - 7 = 0$
22. The equation of the line passing through the point $(1,2)$ with slope $(5,3)$ is ()
- A. $\frac{y-2}{5-1} = \frac{x-1}{3-1}$ B. $\frac{y-2}{3-2} = \frac{x-1}{5-1}$
C. $\frac{y-1}{5-1} = \frac{x-3}{2-3}$ D. $\frac{x-2}{5-2} = \frac{y-3}{1-3}$
23. Let point M be a point on the z axis such that the distance from point M to $A(1,0,2)$ is equal to the distance from point M to $B(1,-3,1)$. Then the coordinates of point M are ()
- A. $(-3,-3,0)$ B. $(0,0,-3)$
C. $(0,-3,-3)$ D. $(0,0,3)$
24. Given point $A(1,0)$ and line $l: x - y + 1 = 0$, the distance from point A to line l is ()
- A. 1 B. 2 C. $\sqrt{2}$ D. $2\sqrt{2}$
25. The radius of circle $x^2 + y^2 + 2x - 4y = 0$ is ()
- A. 3 B. $\sqrt{3}$ C. $\sqrt{5}$ D. 5
26. Given that α is an acute angle, if $\cos \alpha = \frac{1}{3}$, then $\tan \alpha =$ ()
- A. $\frac{\sqrt{2}}{4}$ B. 2 C. $2\sqrt{2}$ D. $\frac{1}{2}$
27. In the geometric sequence $\{a_n\}$, if $a_1 + a_3 = 10, a_4 + a_6 = \frac{5}{4}$, then the general term formula for the sequence $\{a_n\}$ is ()
- A. $a_n = 2^{4-n}$ B. $a_n = 2^{n-4}$ C. $a_n = 2^{n-3}$ D. $a_n = 2^{3-n}$
28. If $a < b < 0$, then which of the following inequalities holds? ()
- A. $\frac{1}{a} < \frac{1}{b}$ B. $-\frac{1}{a} < -\frac{1}{b}$ C. $a^2 < b^2$ D. $ab < b^2$
29. In the spatial Cartesian coordinate system $Oxyz$, the point $M(0,m,0)$ is equidistant from the

points $P(1,0,2)$ and $Q(1,-3,1)$. The value of the real number m is ().

- A. -2 B. -1 C. 1 D. 2

30. Given the function $f(x) = \frac{1}{2} \sin 2x + \sin x$, then $f'(x)$ is ().

- A. An odd function with only a minimum value B. An even function with only a maximum value
C. An even function with both maximum and minimum values D. A function that is neither odd nor even

31. The function $f(x) = \sqrt{x^2 + 5}$ ().

- A. It is an odd function but not an even function B. It is an even function but not an odd function
C. Both an odd function and an even function D. Neither an odd function nor an even function

32. If $\sin \alpha = \frac{1}{5}$, then $\cos 2\alpha =$ ().

- A. $\frac{24}{25}$ B. $\frac{23}{25}$ C. $-\frac{23}{25}$ D. $-\frac{24}{25}$

33. The intersection coordinates of the two lines $l_1 : 2x - y - 1 = 0$ and $l_2 : x + 3y - 11 = 0$ are ().

- A. $(-3 \ 2)$
B. $(-2 \ 3)$
C. $(2 \ 3)$
D. $(3 \ 2)$

34. The equation of the tangent line to the curve $y = \ln x + 1$ at the point where the x-coordinate is 1 is ().

- A. $x + y - 1 = 0$ B. $x + y + 1 = 0$ C. $x + y = 0$ D. $x - y = 0$

35. Given $a = 3 \cdot 2^{0.1}$, $b = \log_2 5$, $c = \log_3 2$, then ().

- A. $b > a > c$ B. $c > b > a$ C. $b > c > a$ D. $a > b > c$

36. Given that $\log_{(18)} a = a$ and $18b = 5$, then $\log_{36} 45 =$ ().

- A. $\frac{2}{a-2}$ B. $\frac{a+b}{2}$ C. $\frac{a+b}{a+2}$ D. $\frac{a+b}{2-a}$

37. Calculate: $\log_2^3 \cdot \log_3^{16} = (\quad)$

- A. 4 B. 8 C. -4 D. -8

38. $\log_2 12 - \log_2 3 = (\quad)$

- A. -2 B. 0 C. $\frac{1}{2}$ D. 2

39. Given the function $f(x) = \sin(2x - \frac{\pi}{2}) (x \in R)$, which of the following conclusions is incorrect? ()

- A. The smallest positive period of the function $f(x)$ is π
 B. The function $f(x)$ is an even function
 C. The graph of $f(x)$ is symmetric with respect to the line $x = \frac{\pi}{4}$
 D. The function $f(x)$ is increasing on the interval $[0, \frac{\pi}{2}]$

40. Regarding the parabola $y = x^2 + 2x - 1$, which of the following statements is incorrect?

()

- A. The vertex coordinates are $(-1, -2)$ B. The axis of symmetry is the line $x = -1$
 C. The opening faces upward D. When $x > -1$, y decreases as x increases

41. Given that the line l passes through the point $M(2, 4)$ and is perpendicular to the line

$x - 2y + 4 = 0$, the equation of the line l is ()

- A. $x - 2y + 1 = 0$ B. $x - 2y - 1 = 0$
 C. $2x - y + 2 = 0$ D. $2x + y - 8 = 0$

42. Given that $\sin \frac{\alpha}{2} - \cos \frac{\alpha}{2} = \frac{1}{5}$, then $\sin \alpha = (\quad)$

- A. $-\frac{24}{25}$ B. $-\frac{12}{25}$ C. $\frac{24}{25}$ D. $\frac{12}{25}$

43. Using the formulas: $\sin(\alpha + \beta) = \sin \alpha \cos \beta + \cos \alpha \sin \beta$ and

$\sin(\alpha - \beta) = \sin \alpha \cos \beta - \cos \alpha \sin \beta$, we obtain: $\sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$.

Simplify the value of $\sin 20^\circ \cos 70^\circ + \sin 10^\circ \sin 50^\circ$: ()

- A. $\frac{1}{4}$ B. $\frac{\sqrt{3}}{2}$ C. $\frac{1}{2}$ D. $\frac{\sqrt{3}}{4}$

44. Which of the following functions is equivalent to the function $y = x^2$? ()

- A. $b = a^2$ B. $y = x^2 - 1$ C. $y = \frac{x^4}{x^2}$ D. $y = x \cdot |x|$

45. Given the complex numbers $z_1 = 6 - 5i$ and $z_2 = 3 + 2i$, where i is the imaginary unit, then

$z_1 + z_2 = (\quad)$

- A. $9 - 3i$ B. $9 + 3i$ C. $9 - 7i$ D. $9 + 7i$

46. Given the arithmetic sequence $\{a_n\}$, where $a_4 + a_9 = 8$, then $S_{12} = (\quad)$

- A. 24 B. 36 C. 48 D. 96

47. Given that the sequence $\{a_n\}$ is an arithmetic sequence and $a_5 + a_7 + a_9 = 18$, the sum of its first 13 terms is $(\quad)$

- A. 45 B. 56 C. 65 D. 78

48. A museum displays the following oracle bone script zodiac animals (see figure). Among them, the horse, ox, sheep, rooster, dog, and pig are the six domesticated animals. If one zodiac animal is randomly selected from each row in the figure, what is the probability that at least one of the three selected animals belongs to the six domesticated animals? $(\quad)$



- A. $\frac{29}{110}$ B. $\frac{3}{4}$ C. $\frac{29}{32}$ D. $\frac{7}{8}$

Reference Answers for 2026 CSCA Mock Mathematics Exam

Question No.	1	2	3	4	5	6	7	8	9	10
Answer	D	C	D	C	D	D	A	A	D	A
Question Number	11	12	13	14	15	16	17	18	19	20
Answer	C	A	B	C	C	B	A	A	A	B
Question Number	21	22	23	24	25	26	27	28	29	30
Answer	A	B	B	C	C	C	A	B	B	C
Question Number	31	32	33	34	35	36	37	38	39	40
Answer	B	B	C	D	A	D	A	D	C	D
Question Number	41	42	43	44	45	46	47	48	44	
Answer	D	C	A	A	A	C	D	C	A	

1. D

【Knowledge Point】Determining the inclusion relationship between two sets

【Analysis】Analyze the relationship between elements in the two given finite sets and another set to obtain the solution.

【Detailed Explanation】Since set B contains only one element 0, and $0 \in A$, we conclude that set B is a subset of set A. Therefore, $B \subseteq A$.

Therefore, $B \subseteq A$.

Therefore, the correct answer is: D

2. C

[Knowledge Point] Solving Univariate Quadratic Inequalities Without Parameters

【Analysis】Solve directly using the method for quadratic inequalities.

【Detailed Solution】The corresponding equation for the inequality is $x^2 - 3x + 2 = 0$,

which is equivalent to $(x-2)(x-1) = 0$.

Solving yields the roots: $x = 2$ or $x = 1$.

∴ The solution set of the inequality $x^2 - 3x + 2 < 0$ is $1 < x < 2$,

i.e., the solution set of the inequality is $\{x | 1 < x < 2\} = (1, 2)$. Therefore, the correct answer is C.

【Key Insight】 This question primarily tests the solution method for quadratic inequalities and is classified as a basic-level problem.

3. D

[Knowledge Point] Domain of a Specific Function

【Analysis】 Based on the properties that the radicand of a quadratic radical must be non-negative and the denominator cannot be zero, we derive the inequality system x . This allows us to determine the domain of the original function.

【Detailed Solution】 Given $\begin{cases} x-1 \geq 0 \\ x-2 \neq 0 \end{cases}$, we solve for $x \geq 1$ and $x \neq 2$. Thus, the domain of the

function $f(x)$ is $[1, 2) \cup (2, +\infty)$.

Therefore, the correct answer is: D.

4. C

[Knowledge Point] Using the general term formula of an arithmetic sequence to find a term

【Explanation】 Calculate using the general term formula for arithmetic sequences.

$$a_8 = a_1 + 7d = 1 + 7 \times 2 = 15 ,$$

Therefore, the correct answer is: C.

【Key Insight】 This question tests the general term formula for arithmetic sequences, classified as an easy problem. The general term formula for an arithmetic sequence is: $a_n = a_1 + (n-1)d$.

5. D

【Knowledge Point】 Numbers and Expressions

【Analysis】 Determine the sign of $x^2 + 2$ to obtain the answer.

【Detailed Explanation】 Since $x^2 + 2 > 0$, the quadrant containing point $P(x^2 + 2, -3)$ is the fourth quadrant.

Therefore, the correct answer is: D.

6. D

【Knowledge Point】 Induction Formulas V and VI

【Analysis】 Using the induction formula simplifies the expression to yield the desired algebraic value.

【Detailed Explanation】 $\cos\left(\frac{\pi}{3}-\alpha\right)=\cos\left[\frac{\pi}{2}-\left(\frac{\pi}{6}+\alpha\right)\right]=\sin\left(\frac{\pi}{6}+\alpha\right)=\frac{4}{5}$.

Therefore, the correct answer is: D.

7. A

【Knowledge Point】 Definition and Determination of Function Oddness/Evenness

【Analysis】 First determine whether the domain is symmetric about the origin, then apply the definition of an even function.

【Detailed Explanation】 For the function in C, its domain is $\{x|x \neq 1\}$. This domain is not symmetric about the origin, so $y = \frac{1}{x-1}$ is not an odd function.

The functions in A, B, and D all have the domain R , which is symmetric about the origin.

For function A, let $f(x) = x^3 + x$. Then $f(-x) = -x^3 - x = -f(x)$ holds, so function A is an odd function.

For the function in B, let $g(x) = x^3 + x^2$. Then $g(-1) = 0, g(1) = 2$,

Since $g(-1) \neq -g(1)$, the functions in B are not odd functions.

For the function in D, let $h(x) = x^2 + 6$. Then $h(0) = 6 \neq 0$. Therefore, the function in D is not an odd function.

Therefore, the correct answer is: A.

[Key Insight] This question tests the determination of odd functions. Note that one should first consider whether the domain is symmetric about the origin, then judge based on the definition. To prove a function is not odd, it suffices to provide a counterexample.

8. A

【Knowledge Point】 Finding Inverse Functions

【Analysis】 Apply the method for finding inverse functions, utilizing the conversion formula between exponential and logarithmic expressions.

【Detailed Solution】 Given the function $y = \log_3 x$, we obtain $x = 3^y$.

Therefore, the inverse function of $y = \log_3 x$ is $y = 3^x$.

Therefore, the correct answer is: A.

9. D

[Knowledge Point] Directly determining the monotonicity of a function based on its analytical expression, determining the monotonicity of exponential functions, investigating the monotonicity of logarithmic functions, determining the monotonicity of functions containing $\tan x$

【Analysis】 The result can be directly determined based on the monotonicity properties of exponential functions, inverse proportional functions, tangent functions, and logarithmic functions.

【Detailed Explanation】 For A, $y = \left(\frac{1}{2}\right)^x$ is monotonically decreasing on $\mathbf{R}$, so A is incorrect;

For B, $y = \frac{1}{x}$ is monotonically decreasing on $(-\infty, 0)$ and $(0, +\infty)$, so B is incorrect;

For C, $y = \tan x$ is monotonically increasing on $\left(-\frac{\pi}{2} + k\pi, \frac{\pi}{2} + k\pi\right) (k \in \mathbf{Z})$, so C is incorrect;

For D, $y = \log_{\sqrt{2}} x$ is monotonically increasing on $(0, +\infty)$, so D is correct.

Therefore, the correct answer is: D.

10. A

[Knowledge Point] Formulas on Coordinate Axes

【Explanation】 Given that point M lies in the fourth quadrant, its x-coordinate is greater than 0 and its y-coordinate is less than 0. Therefore, the coordinates are determined based on the distance to the coordinate axes.

【Detailed Explanation】 Since *point M* lies in the fourth quadrant, its x-coordinate is positive and y-coordinate is negative.

and since *point M* is 6 units from the x-axis and 4 units from the y-axis, the coordinates of *point M* are (4, -6).

Therefore, the correct answer is: A

11. C

[Knowledge Point] Fractional Inequalities

[Analysis] By rearranging and simplifying the fraction inequality, we obtain $\begin{cases} (x-1)(x-4) \leq 0 \\ x-1 \neq 0 \end{cases}$,

which can be solved.

【Detailed Solution】 From $\frac{x+2}{x-1} \geq 2$,

we obtain: $\frac{x+2}{x-1} - 2 \geq 0$,

which implies $\frac{-x+4}{x-1} \geq 0$,

which simplifies to $\begin{cases} (x-1)(x-4) \leq 0 \\ x-1 \neq 0 \end{cases}$,

Solving yields: $1 < x \leq 4$,

Therefore, the correct answer is: C

12. A

[Knowledge Point] Calculate using the properties of arithmetic sequences

[Analysis] Solve using properties of arithmetic sequences.

【Detailed Solution】 By the properties of arithmetic sequences, we have: $a_3 + a_7 = 2a_5$.

Therefore, $a_5 = 5$.

Therefore, the correct answer is: A.

13. B

【Knowledge Point】 Application of the slope formula

【Analysis】 The answer can be derived from the relationship between slope and angle of inclination.

【Detailed Explanation】 Let the angle of inclination be α . Then, $\tan \alpha = 1$. Since $0^\circ \leq \alpha < 180^\circ$,

Then $\alpha = 45^\circ$.

Therefore, the correct answer is: B.

14. C

[Knowledge Point] Comparing the sizes of exponential powers and logarithmic expressions

【Analysis】 The result can be determined based on the monotonicity of exponential and logarithmic functions.

【Detailed Explanation】 For A, since the base is $\frac{3}{4} < 1$, the function decreases as the exponent increases. Furthermore, $\frac{2}{3} < \frac{3}{4}$, so $\left(\frac{3}{4}\right)^{\frac{2}{3}} > \left(\frac{3}{4}\right)^{\frac{3}{4}}$. Therefore, option A is incorrect.

For B, $\log_4 12 = \frac{1}{2} \log_2 12 = \log_2 \sqrt{12} = \log_2 2\sqrt{3}$. Since the base is $2 > 1$, the function increases as the argument increases. Furthermore, $5 > 2\sqrt{3}$ implies $\log_2 5 > \log_2 2\sqrt{3}$. Therefore, option B is incorrect.

For C, since $\log_7 3 > \log_7 \sqrt{7} = \frac{1}{2}$ and $\frac{1}{2} = \frac{\sqrt{5}}{2\sqrt{5}} > \frac{\sqrt{5}}{5}$, then $\log_7 3 > \frac{\sqrt{5}}{5}$. Therefore, option C is correct.

For D, since $\left[(\sqrt{2})^{3.9}\right]^2 = 2^{3.9}$ and $(3.9)^2$, the function $2^x, x^2$ has two intersection points: when $x = 2, x = 4$ and

2^x the growth rate exceeds that of x^2 . At $(0, 2)$, $2^x > x^2$ holds, and at $(2, 4)$, $2^x < x^2$ holds.

On $(4, +\infty)$, $2^x > x^2$ holds, so $2^{3.9} < (3.9)^2$, i.e., $(\sqrt{2})^{3.9} < 3.9$. Therefore, option D is incorrect.

Therefore, the correct answer is: C.

15. C

[Knowledge Point] Comparing the sizes of logarithmic expressions, comparing the sizes of exponential powers

[Analysis] Based on the monotonicity of exponential and logarithmic functions, evaluate each term sequentially to solve.

【Detailed Explanation】 For option A, since the function $y = 3^x$ is monotonically increasing, we have $3^{0.8} > 3^{0.7}$. Therefore, A is incorrect.

For option B, since the function $y = 0.75^x$ is monotonically decreasing, we have $0.75^{-0.1} > 0.75^{0.1}$, so B is incorrect;

For C, since the function $y = \log_2 x$ is monotonically increasing, we have $\log_2 3.4 < \log_2 8.5$, so C is correct;

For option D, since the function $y = \log_{0.3} x$ is monotonically decreasing, we have

$\log_{0.3} 1.8 > \log_{0.3} 2.7$, so D is incorrect.

Therefore, the correct answer is: C.

16. B

【Explanation】 From the coordinates of the point on the terminal side, we have $\tan \alpha = -3$.

Using the double-angle formula for tangent, we obtain the value of $\tan 2\alpha$.

【Detailed Explanation】 Given that point $P(-1, 3)$ lies on the terminal side of angle α , we have:

$\tan \alpha = -3$.

Therefore, by the double-angle formula for tangent, $\tan 2\alpha = \frac{2 \tan \alpha}{1 - \tan^2 \alpha} = \frac{3}{4}$,

Therefore, the correct answer is: B

【Key Insight】 This problem tests the application of the double-angle formula for tangent to calculate values. It involves the comprehensive use of coordinates of points on the terminal side to find the sine value of an angle, then applying the double-angle formula for calculation. It is classified as a straightforward problem.

17. A

[Knowledge Point] Calculation of the common term formula for arithmetic sequences

【Analysis】 Solve using the fundamental calculation of arithmetic sequence constants.

【Detailed Solution】 Let the common difference of the arithmetic sequence be d . From

$a_1 = 1, a_3 = 3$, we obtain $a_3 = a_1 + 2d$.

Thus, the common difference is $d = 1$.

Thus, $a_{2024} = a_1 + 2023d = 2024$,

Therefore, the correct answer is: A

18. A

【Knowledge Point】 Conversion between the general equation and standard equation of a circle; finding the equation of a circle from its center (or radius)

【Analysis】 First derive the standard form of the circle based on the problem statement, then convert it to the general form.

【Detailed Solution】 Given that the radius of circle C is 2, the equation of circle C is

$$(x-1)^2 + (y-2)^2 = 4 \quad .$$

which is equivalent to $x^2 + y^2 - 2x - 4y + 1 = 0$. Therefore, option A is correct.

Therefore, the correct answer is: A.

19. A

[Knowledge Point] Determining the center and radius of a circle from its standard equation

[Analysis] The center coordinates can be obtained directly from the standard form of the circle equation.

【Detailed Explanation】 The center coordinates of the circle $\because (x-a)^2 + (y-b)^2 = r^2$ are (a, b) ;

$\therefore (x+1)^2 + (y-2)^2 = 4$ The center coordinates are $((-1, 2))$;

Therefore, the correct answer is: A.

20. B

[Knowledge Point] Deriving the equation from two perpendicular lines, point-slope form of a line equation, and analysis

[Analysis] The slope of the line can be obtained from its normal vector, and the result can be derived from the point-slope form equation of the line.

[Detailed Solution] The normal vector of the line $\because$ is $(2, 1)$. The slope of the line $\therefore$ is $k = -2$.

$\therefore$ The equation of the line is $y-1 = -2(x-1)$, which is equivalent to $2x + y - 3 = 0$.

Therefore, the correct answer is: B

21. A

[Knowledge Point] Point-Slope Form of a Line and Analysis

【Analysis】 Solve using the point-slope form of the line equation based on the given conditions.

【Detailed Solution】 The equation of the line passing through point $(3, 1)$ with slope $\frac{1}{2}$ is

$$y-1 = \frac{1}{2}(x-3) \quad , \text{ which is equivalent to } x-2y-1=0 \quad .$$

Therefore, the correct answer is: A.

22. B

【Knowledge Point】 Two-Point Form of Line Equation and Analysis

【Analysis】 Directly use the two-point form to write the line equation.

【Detailed Explanation】 Since the line passes through the points $(1, 2)$ and $(5, 3)$, the equation is $\frac{y-2}{3-2} = \frac{x-1}{5-1}$.

Therefore, the correct answer is: B.

23. B

[Knowledge Point] Application of the Spatial Distance Formula

【Analysis】 Define the coordinates of point M . Solve the equation based on the given conditions.

【Detailed Solution】 Let $M(0, 0, m)$. Then $\sqrt{1^2 + 0 + (m-2)^2} = \sqrt{1^2 + (-3)^2 + (m-1)^2}$,

Solving yields $m = -3$, so the coordinates of M are $(0, 0, -3)$.

Therefore, the correct answer is: B

24. C

[Knowledge Point] Finding the distance from a point to a line

[Analysis] Calculate using the distance formula from a point to a line.

【Detailed Solution】 Solution: Point $A(1, 0)$, line $l: x - y + 1 = 0$. The distance from point A to line l is $\frac{|1-0+1|}{\sqrt{1^2+1^2}} = \sqrt{2}$,

Therefore, the correct answer is: C.

【Key Insight】 The distance from point $P(x_0, y_0)$ to line $Ax + By + C = 0$ is

$$d = \frac{|Ax_0 + By_0 + C|}{\sqrt{A^2 + B^2}}.$$

25. C

[Knowledge Point] Equation of a Circle

【Detailed Explanation】 Question Analysis: Transform $x^2 + y^2 + 2x - 4y = 0$ into

$$(x+1)^2 + (y-2)^2 = 5 \therefore r^2 = 5 \therefore r = \sqrt{5}$$

Test Point: Equation of a Circle

26. C

【Knowledge Point】 Finding Tangent (Chord) from Given Chord (Tangent), Finding Sine

(Cosine) from Given Cosine (Sine)

【Analysis】 Based on trigonometric identities for the same angle, given the cosine value of an angle, find the tangent value.

【Detailed Solution】 Given that α is an acute angle, then $\sin \alpha = \sqrt{1 - \cos^2 \alpha} = \frac{2\sqrt{2}}{3}$,

$$\text{Therefore, } \tan \alpha = \frac{\sin \alpha}{\cos \alpha} = \frac{\frac{2\sqrt{2}}{3}}{\frac{1}{3}} = 2\sqrt{2}.$$

Therefore, the correct answer is: C.

27. A

【Knowledge Point】 Write the general term formula for a geometric sequence and calculate the basic quantity of the general term formula for a geometric sequence.

【Analysis】 Substitute into the two equations using the general term formula for a geometric sequence to determine the first term and common ratio, then evaluate the options.

【Detailed Solution】 In the geometric sequence $\{a_n\}$, $a_1 + a_3 = 10$, $a_4 + a_6 = \frac{5}{4}$,

$$\text{From the general term formula for a geometric sequence, we obtain } \begin{cases} a_1 + a_1 q^2 = 10 \\ a_1 q^3 + a_1 q^5 = \frac{5}{4} \end{cases}$$

Dividing the two equations yields $q^3 = \frac{1}{8}$, which implies $q = \frac{1}{2}$.

Substituting yields $a_1 = 8 = 2^3$,

Therefore, using the general term formula for a geometric sequence, we obtain

$$a_n = 2^3 \cdot \left(\frac{1}{2}\right)^{n-1} = 2^{4-n}.$$

Therefore, the correct answer is: A.

【Key Insight】 This question tests the basic application of the general term formula for geometric sequences, representing a foundational problem.

28. B

[Knowledge Point] Determine the validity of given inequalities based on conditions; compare the magnitude of numbers (expressions) using properties of inequalities.

【Analysis】 Apply the properties of inequalities to analyze each option sequentially to determine the correct answer.

【Detailed Explanation】 Option A: Since $a < b < 0$, then $0 > \frac{1}{a} > \frac{1}{b}$, thus A is incorrect.

Option B: Since $0 > \frac{1}{a} > \frac{1}{b}$, then $-\frac{1}{a} < -\frac{1}{b}$, therefore B is correct;

Option C: Since $a < b < 0$, then $a^2 > b^2$, therefore C is incorrect;

Option D: Since $a < b < 0$, then $ab > b^2$, therefore D is incorrect.

Therefore, the correct answer is: B

29. B

[Knowledge Point] Application of the Spatial Distance Formula

[Analysis] Based on the distance formula between two points in a spatial Cartesian coordinate system, establish the equation for m and solve it.

[Detailed Solution]: $M(0, m, 0), P(1, 0, 2), Q(1, -3, 1), |MP| = |MQ|$,

$$\therefore 1^2 + (-m)^2 + 2^2 = 1^2 + (-3 - m)^2 + 1^2 \text{ Solving yields } m = -1.$$

Therefore, the correct answer is: B.

【Key Insight】 This question tests basic operations in a spatial Cartesian coordinate system and is considered foundational.

30. C

【Knowledge Point】 Finding the maximum/minimum value of a function (without parameters) using derivatives; definition and determination of function oddness/evenness

【Detailed Solution】 Problem Analysis: First differentiate to transform into a quadratic function, then use the monotonicity of trigonometric functions to find the maximum value. Finally, determine the odd/even property using the definition of function oddness/evenness.

Solution: Since $f(x) = \frac{1}{2}\sin 2x + \sin x$,

$$\therefore f'(x) = \cos 2x + \cos x = 2\cos^2 x + \cos x - 1 = 2\left(\cos x + \frac{1}{4}\right)^2 - \frac{9}{8}. \text{ When } \cos x = -\frac{1}{4}, f'(x)$$

attains its minimum value $-\frac{9}{8}$; when $\cos x = 1$, $f(x)$ attains its maximum value 2.

Moreover, $f(-x) = f(x)$. Thus, $f(x)$ is an even function possessing both a maximum and a minimum value.

Therefore, the correct answer is C.

Commentary: Mastery of the derivatives of composite functions, the maximum and minimum

values of quadratic functions, the monotonicity of trigonometric functions, and the evenness and oddness of functions is key to solving this problem.

31. B

[Knowledge Point] Definition and Determination of Function Parity

[Analysis] First determine the domain of the function, then apply the definition of even/odd functions to select the correct option.

[Detailed Explanation] The domain of the function $f(x) = \sqrt{x^2 + 5}$ is $\mathbb{R}$, symmetric about the origin.

$$\text{and since } f(-x) = \sqrt{(-x)^2 + 5} = \sqrt{x^2 + 5} = f(x),$$

which implies $f(-x) = f(x)$,

Therefore, the function is even.

Therefore, the correct answer is: B

32. B

[Knowledge Point] Cosine formula for double angle

[Analysis] Solve using the cosine formula for double angles.

【Detailed Solution】Solution: Since $\sin \alpha = \frac{1}{5}$,

$$\text{it follows that } \cos 2\alpha = 1 - 2\sin^2 \alpha = 1 - 2 \times \left(\frac{1}{5}\right)^2 = \frac{23}{25},$$

Therefore, the correct answer is: B

33. C

【Knowledge Point】Finding the coordinates of the intersection point of two lines

【Analysis】Solve the system of equations by combining the equations of both lines.

【Detailed Solution】Given line $l_1: 2x - y - 1 = 0$, and line $l_2: x + 3y - 11 = 0$,

$$\text{From } \begin{cases} 2x - y - 1 = 0 \\ x + 3y - 11 = 0 \end{cases}, \text{ we solve for: } \begin{cases} x = 2 \\ y = 3 \end{cases},$$

Therefore, the intersection point coordinates of lines l_1 and l_2 are $(2, 3)$.

Therefore, the correct answer is: C.

34. D

[Knowledge Point] Derivative formulas for basic elementary functions; finding the equation (slope) of a tangent line at a point on a curve

【Analysis】 Solve using the geometric meaning of derivatives.

【Detailed Solution】 Solution: Since $y = f(x) = \ln x + 1$, then $f'(x) = \frac{1}{x}$,

thus the slope of the tangent line is $k = f'(1) = 1$.

and since $f(1) = \ln 1 + 1 = 1$, the coordinates of the tangent point are $(1, 1)$.

Thus the tangent line equation is $y - 1 = 1(x - 1)$, i.e., $x - y = 0$.

Therefore, the correct answer is: D.

35. A

[Knowledge Point] Comparing the sizes of logarithmic expressions, comparing the sizes of exponential powers

【Analysis】 The result follows directly from the monotonicity properties of exponential and logarithmic functions.

【Detailed Explanation】 $1 = 3.2^0 < 3.2^{0.1} < 3.2^{0.5} < 2 \Rightarrow 1 < a < 2$

$\log_2 5 > \log_2 4 = 2 \Rightarrow b > 2$

$0 = \log_3 1 < \log_3 2 < \log_3 3 = 1 \Rightarrow 0 < c < 1$

Therefore $b > a > c$

Therefore, the correct answer is: A

36. D

【Key Point】 Application of logarithmic properties, simplification using change-of-base formula

【Analysis】 Convert the exponential expression to logarithmic form, use the change-of-base formula to convert to the same base, then transform using logarithmic operation rules.

【Detailed Solution】 From $18^b = 5$, we obtain $\log_{18} 5 = b$.

$$\log_{36} 45 = \frac{\log_{18} 45}{\log_{18} 36} = \frac{\log_{18} (5 \times 9)}{\log_{18} \frac{18^2}{9}} = \frac{\log_{18} 5 + \log_{18} 9}{2 - \log_{18} 9} = \frac{b + a}{2 - a}.$$

Therefore, the correct answer is: D.

【Key Insight】 This question tests the change-of-base formula for logarithms and logarithmic operation rules, constituting a foundational problem.

37. A

【Analysis】 By applying logarithmic function operations, the result can be simplified.

【Detailed Solution】 By applying logarithmic function operations, we find that

$$\begin{aligned}\log_2 3 \cdot \log_3 16 &= \log_2 3 \cdot \log_3 2^4 \\ &= \log_2 3 \cdot 4 \log_3 2 = 4 \log_2 3 \cdot \log_3 2 = 4.\end{aligned}$$

Therefore, the correct answer is: A.

【Key Insight】 This question tests logarithmic operations and is considered a foundational problem.

38. D

【Knowledge Point】 Logarithmic Functions

【Detailed Solution】 Question Analysis: Directly simplify and evaluate using logarithmic properties.

$$\text{Solution: } \log_2^{12} - \log_2^3 = \log_2 \frac{12}{3} = \log_2 4 = \log_2 2^2 = 2.$$

Therefore, the correct answer is D.

Test Point: Properties of logarithmic operations.

39. C

【Knowledge Point】 Graphs and Properties of Trigonometric Functions

【Detailed Solution】 Analysis: The original function simplifies using the induction formula to:

$$f(x) = \sin\left(2x - \frac{\pi}{2}\right) = -\cos 2x. \text{ This function is an even function with the smallest positive}$$

period π , so A and B are correct. The axis of symmetry is obtained from: $2x = k\pi (k \in \mathbb{Z})$:

$x = \frac{k\pi}{2} (k \in \mathbb{Z})$. Clearly, for any integer values of k , $x \neq \frac{\pi}{4}$ holds, so C is incorrect. The answer is C.

Key points: 1. Induction formula; 2. Properties of trigonometric functions.

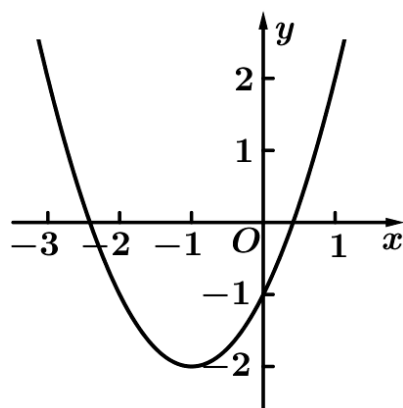
40. D

【Knowledge Point】 Analysis and Judgment of Quadratic Function Graphs

【Detailed Solution】 As shown in the figure, the parabola is defined by

$$y = x^2 + 2x - 1 = (x+1)^2 - 2. \text{ The vertex coordinates are } (-1, -2), \text{ so A is correct. The axis of}$$

symmetry is the line $x = -1$, so B is correct. Since $a = 1 > 0$, the parabola opens upward, so C is correct. When $x > -1$, y increases as x increases, so D is incorrect.



[Essential Problem-Solving Skills]

1. Properties of the quadratic function $y = ax^2 + bx + c$:

Sign of a	$a > 0$	$a < 0$
Graph		
Direction of Opening	Upward	Downward
Symmetry axis	$x = -\frac{b}{2a}$	$x = -\frac{b}{2a}$
Vertex Coordinates	$\left(-\frac{b}{2a}, \frac{4ac - b^2}{4a}\right)$	$\left(-\frac{b}{2a}, \frac{4ac - b^2}{4a}\right)$
Increasing/decreasing property	When $x < -\frac{b}{2a}$, y decreases as x increases; when $x > -\frac{b}{2a}$, y increases as x increases	When $x < -\frac{b}{2a}$, y increases as x increases; when $x > -\frac{b}{2a}$, y decreases as x increases

Extremum	When $x = -\frac{b}{2a}$, $y_{\text{vertex}} = \frac{4ac - b^2}{4a}$	When $x = -\frac{b}{2a}$, $y_{\text{vertex}} = \frac{4ac - b^2}{4a}$
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2. If the parabola $y = ax^2 + bx + c$ intersects the x-axis at a point with x-coordinate x_0 , then when $x = x_0$, the function value is 0. Therefore, $x = x_0$ is a root of the equation $ax^2 + bx + c = 0$.

3. The graph of the quadratic function $y = ax^2 + bx + c$ has three possible relationships with the x-axis, corresponding to the three cases of the roots of the quadratic equation $ax^2 + bx + c = 0$.

(1) When $b^2 - 4ac < 0$, the graph of $y = ax^2 + bx + c$ has no intersection with the x-axis, and the quadratic equation $ax^2 + bx + c = 0$ has no real roots.

(2) When $b^2 - 4ac = 0$, the graph of the quadratic function $y = ax^2 + bx + c$ intersects the x-axis at one point, and the quadratic equation $ax^2 + bx + c = 0$ has two equal real roots.

(3) When $b^2 - 4ac > 0$, the graph of the quadratic function $y = ax^2 + bx + c$ intersects the x-axis at two points, and the quadratic equation $ax^2 + bx + c = 0$ has two unequal real roots.

41. D

[Knowledge Point] Deriving the equation of a line from the perpendicular relationship between two lines

【Analysis】 Based on the perpendicular relationship, establish the equation of line l . Substitute into $M(2, 4)$ to find the solution.

【Detailed Solution】 Let the equation of line l be $2x + y + C = 0$.

Substituting $M(2, 4)$ into $2x + y + C = 0$ yields: $4 + 4 + C = 0$. Therefore, $C = -8$.

Therefore, the equation of the line l is $2x + y - 8 = 0$.

Therefore, the correct answer is: D

42. C

[Knowledge Point] Double-angle sine formula, relationship between $\sin\alpha \pm \cos\alpha$ and $\sin\alpha \cdot \cos\alpha$

[Analysis] Use the squaring method to find the correct answer.

【Detailed Solution】 Squaring both sides of $\sin \frac{\alpha}{2} - \cos \frac{\alpha}{2} = \frac{1}{5}$ yields

$$\sin^2 \frac{\alpha}{2} - 2\sin \frac{\alpha}{2} \cos \frac{\alpha}{2} + \cos^2 \frac{\alpha}{2} = \frac{1}{25},$$

$$\text{which implies } 1 - \sin \alpha = \frac{1}{25}, \sin \alpha = \frac{24}{25}.$$

Therefore, the correct answer is: C

43. A

【Knowledge Point】 Simplifying and evaluating trigonometric expressions using the sine formulas for sum and difference angles; trigonometric function simplification and evaluation—induction formulas

【Analysis】 Calculate and simplify based on the given formula to find the value.

【Detailed Solution】 From $\sin \alpha \cos \beta = \frac{1}{2} [\sin(\alpha + \beta) + \sin(\alpha - \beta)]$, we obtain:

$$\sin 20^\circ \cos 70^\circ = \frac{1}{2} [\sin(20^\circ + 70^\circ) + \sin(20^\circ - 70^\circ)] = \frac{1}{2} \sin 90^\circ + \frac{1}{2} \sin(-50^\circ),$$

$$\sin 10^\circ \sin 50^\circ = \sin 10^\circ \sin(90^\circ - 50^\circ) = \sin 10^\circ \cos 40^\circ$$

$$= \frac{1}{2} [\sin(10^\circ + 40^\circ) + \sin(10^\circ - 40^\circ)] = \frac{1}{2} \sin 50^\circ + \frac{1}{2} \sin(-30^\circ),$$

$$\text{Therefore, } \sin 20^\circ \cos 70^\circ + \sin 10^\circ \sin 50^\circ = \frac{1}{2} \sin 90^\circ + \frac{1}{2} \sin(-50^\circ) + \frac{1}{2} \sin 50^\circ + \frac{1}{2} \sin(-30^\circ)$$

$$= \frac{1}{2} \sin 90^\circ - \frac{1}{2} \sin 50^\circ + \frac{1}{2} \sin 50^\circ - \frac{1}{2} \sin 30^\circ = \frac{1}{2} - \frac{1}{4} = \frac{1}{4},$$

Therefore, the correct answer is: A.

44. A

[Knowledge Point] Determining whether two functions are equal

[Analysis] Based on the definition of an identical function, evaluate each option by examining both the domain and the correspondence relationship.

【Detailed Explanation】 The domain of $y = x^2$ is $\mathbb{R}$.

For option A, both the domain and correspondence of $b = a^2$ match those of $y = x^2$,

confirming they are the same function. Thus, A is correct.

For option B, the correspondence between $y = x^2 - 1$ and $y = x^2$ differs, indicating they are not the same function, thus option B is incorrect;

For C, the domain of $y = \frac{x^4}{x^2}$ is $\{x | x \neq 0\}$, while the domain of $y = x^2$ is $\mathbb{R}$.

Therefore, they are not the same function, making C incorrect;

For D, regarding the function $y = x \cdot |x|$, when $x < 0$, $y = -x^2$. This differs from the correspondence $y = x^2$, so they are not the same function. Therefore, D is incorrect.

Therefore, the correct answer is: A

45. A

[Knowledge Point] Algebraic operations for complex number addition and subtraction

[Analysis] Solve using complex addition operations.

【Detailed Explanation】 Since $z_1 = 6 - 5i$ and $z_2 = 3 + 2i$, then

$$z_1 + z_2 = (6 - 5i) + (3 + 2i) = 9 - 3i.$$

Therefore, the correct answer is: A.

46. C

[Knowledge Point] Finding the sum of the first n terms of an arithmetic sequence; calculating using properties of arithmetic sequences

【Analysis】 Using the property of the general term of an arithmetic sequence

$$a_4 + a_9 = a_1 + a_{12}, \text{ we can find } S_{12}.$$

【Detailed Explanation】 In the arithmetic sequence $\{a_n\}$, $a_4 + a_9 = 8$,

$$\text{then } S_{12} = \frac{12(a_1 + a_{12})}{2} = \frac{12(a_4 + a_9)}{2} = 6 \times 8 = 48.$$

Therefore, the correct answer is: C.

47. D

[Knowledge Point] Calculating the sum of the first n terms and the general term formula of an arithmetic sequence

【Analysis】 By the arithmetic mean of the sequence, $a_7 = 6$. Using the sum formula and properties, we find $S_{13} = 13a_7$.

【Detailed Solution】 Since in the arithmetic sequence $\{a_n\}$, $a_5 + a_7 + a_9 = 18$, we have $a_5 + a_7 + a_9 = 3a_7 = 18$.

Solving for a_7 yields $a_7 = 6$.

Therefore, the sum of the first 13 terms of this sequence is:

$$S_{13} = \frac{13}{2} \times (a_1 + a_{13}) = 13a_7 = 13 \times 6 = 78.$$

Therefore, the correct answer is D.

[Key Insight] This problem tests the sum of the first n terms of an arithmetic sequence. Utilizing the properties of arithmetic sequences and the formula for S_n is key to solving it, making this a foundational question.

48. C

[Knowledge Point] Calculating probabilities in classical probability models, combinatorial counting problems in real-world scenarios

【Analysis】 Calculate the probability by considering two scenarios: selecting a monkey in the third row and not selecting a monkey in the third row.

【Detailed Solution】 If a monkey is chosen in the third row, at least one of the first two rows must select a zodiac animal from the six domestic animals.

yielding $C_1^1 C_4^1 + C_3^1 C_2^1 = 10$ possible selections.

If no monkey is chosen in the third row, there are $C_3^1 C_4^1 C_4^1 = 48$ possible selections,

Thus, the probability is $\frac{10 + 48}{C_4^1 C_4^1 C_4^1} = \frac{29}{32}$.

Therefore, the correct answer is: C.